

L&T EduTech Project Submission

AI-Based Road Surface Defect Monitoring System

Day 4: UAV Aerial Model Training & Analytics Schema

Aryan Jaljith CB.EN.U4EEE23105

Harish R. CB.EN.U4EEE23112

Karthik K. CB.EN.U4EEE23116

Tanushri Ravish CB.EN.U4EEE23035

Internal Mentor: Mr. Sathya Vignesh

Department of Electrical & Electronics Engineering
Amrita Vishwa Vidyapeetham, Coimbatore

June 2026

4.1 Aerial Dataset Training Challenges

Day 4 centered on training the second half of the architecture: the UAV Drone model. The top-down aerial perspective introduces the classic computer vision problem of Small Object Detection (SOD). From an altitude of 20 to 50 meters, the Ground Sample Distance (GSD) increases significantly, meaning defects occupy far fewer pixels compared to the dash-cam footage. Furthermore, critical contextual cues, such as the horizon line or perspective scaling, are entirely absent. The second YOLOv8 instance was initialized and trained exclusively on the augmented UAV dataset to adapt its receptive fields to these flat, top-down geometries.

4.2 Drone Model Performance Evaluation

Despite the reduced spatial resolution of anomalies, the drone model's performance metrics were highly competitive:

- **Aggregate Metrics:** The model achieved an overall mAP@0.5 of **0.883**. While slightly lower than the vehicle model, this is considered state-of-the-art for aerial anomaly detection.
- **Infrastructural Recognition:** Manholes (D50) were detected with an outstanding 0.972 precision. This validates the network's ability to utilize geometric edge-detection to isolate intentional infrastructure from organic road degradation, effectively preventing a massive source of false positives.
- **Pothole vs. Crack Discrepancy:** Severe Potholes (D20) scored a near-perfect 0.995 precision. However, Severe Cracks (0.751) proved more challenging. The loss of depth perception in 2D aerial mapping causes severe cracks to occasionally blend with normal asphalt seams or tar lines, leading to a slight drop in recall.

The drone model weights (`best_drone.pt`) were finalized and aligned alongside the vehicle model, officially completing the machine learning phase of the project.

4.3 Data Pipeline Bridging & JSON Schema Design

With both AI models active and accurate, the focus shifted to Milestone 3: constructing the bridging architecture that translates pixel bounding boxes into actionable database rows. We developed a Python script utilizing the `ultralytics` inference API combined with `pandas` and `gsread`.

The script takes an input image, runs the inference, loops through the detected bounding boxes, and calculates a custom severity score. Crucially, it attaches simulated GPS coordinates to the detection. The output schema was strictly defined to ensure seamless integration with downstream BI tools:

Field Name	Data Type	Description
Record_ID	Integer	Unique sequential identifier based on epoch time.
Timestamp	ISO 8601	UTC time of detection for seasonal tracking.
Latitude / Long	Float	High-precision GPS mapping coordinates.
Defect_Class	String	Enum class (e.g., Pothole_Severe, Crack_Mild).
Priority_Level	String	Categorical output (CRITICAL, HIGH, MEDIUM).
Source	String	Tracks the originating model (Road Camera vs. Drone).

This script automatically flattens the multi-object detections into standard tabular rows and pushes them via an authenticated API directly to a live Google Sheet, acting as our cloud database ledger.